

Mark Scheme (Results)

Summer 2026 (Predicted)

Pearson Edexcel GCE

In Economics B (9EB0)

Paper 03: The Economic Environment and Business

PREDICTED PRACTICE PAPER — ADVANCED

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General Marking Guidance

- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme not according to their perception of where the grade boundaries may lie.
- There is no ceiling on achievement. All marks on the mark scheme should be used appropriately.
- All the marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Examiners should also be prepared to award zero marks if the candidate's response is not worthy of credit according to the mark scheme.
- Where some judgement is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt regarding the application of the mark scheme to a candidate's response, the team leader must be consulted.
- Crossed out work should be marked UNLESS the candidate has replaced it with an alternative response.

Section A — Question 1

Case-study context: the global semiconductor industry, the CHIPS Act, and the US–China technology decoupling.

Question Number	With reference to Extract A and using a suitable market diagram, analyse why the global market for leading-edge semiconductor manufacturing is dominated by such a small number of firms.	Mark
	<i>Indicative content</i>	
1(a)	Knowledge/understanding 2, Application 2, Analysis 2, Evaluation 2 • Diagram (required): LRAC falling throughout the relevant output range showing that a small number of firms can achieve minimum efficient scale (natural oligopoly); or a high-barriers-to-entry oligopoly diagram. Axes and curves must be correctly labelled. • Extract A — a single leading-edge fab costs US\$15–30 billion and takes 3–5 years to bring on-line. TSMC alone accounts for 60% of foundry output and 90%+ at ≤5nm. • Reasons for concentration: (i) extreme sunk fixed cost — fabs cannot be easily repurposed; (ii) minimum efficient scale is very large relative to demand — LRAC falls over most of the relevant range; (iii) learning-curve effects — yield on advanced nodes takes years to mature; (iv) agglomeration economies in Hsinchu/elsewhere — specialised suppliers, labour pool, tacit knowledge; (v) patents and IP entrench incumbents; (vi) capital-market access — only a few firms can raise the required capital. • Upstream dependencies reinforce concentration: ASML is the sole global producer of EUV lithography; EDA software is a three-firm oligopoly. • Evaluation: depends on (i) pace of technological change — new architectures (chiplets, advanced packaging) may create entry windows; (ii) government subsidy — CHIPS Act and China's Big Fund artificially lower effective entry cost; (iii) node maturity — older (28nm+) nodes are far less concentrated. • Judgement: structural forces strongly favour continued concentration at the leading edge, though policy may slow the trend.	(8)

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Level 3	6–8	Accurate knowledge and understanding, supported throughout by use of relevant evidence which is well chosen. Arguments are developed, using logical, coherent chains of reasoning. A balanced awareness of competing arguments.

Demonstrating application (AO2) in responses

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Question Number	With reference to Extracts B and C and Figures 1 and 2, discuss the macroeconomic consequences for the United States of the concentration of semiconductor production in East Asia.	Mark
	<i>Indicative content</i>	
1(b)	Knowledge/understanding 2, Application 2, Analysis 3, Evaluation 3 • Knowledge: macro indicators — GDP, inflation, unemployment, current account, national security externality. Resilience-vs-efficiency trade-off. • Figure 1 — US share of global production fell from 37% (1990) to 11% (2024); Figure 2 — US announced \$223bn capex. Extract C — Taiwan concentration; Rhodium-estimated multi-trillion-dollar global GDP hit from disruption. • Negative consequences: (i) supply-chain vulnerability — Taiwan contingency could cut US GDP by 3–5% in severe scenarios; (ii) national-security exposure — defence systems depend on leading-edge chips; (iii) ICT trade deficit; (iv) reduced dynamic comparative advantage — loss of tacit manufacturing knowledge; (v) single-point-of-failure inflation vulnerability. • Positive/mitigating: (i) US retains dominance in highest-value segments — design (Nvidia, Qualcomm), EDA, equipment (Applied Materials, Lam), IP; US share of value captured is far larger than share of production; (ii) offshoring has historically produced lower consumer prices; (iii) CHIPS Act capex (\$223bn) is already reshoring some capacity. • Evaluation: depends on (i) probability of actual disruption — low but non-negligible; (ii) time horizon — short-run exposure high, long-run depends on reshoring success; (iii) counterfactual cost of self-sufficiency — significant fiscal cost and likely higher consumer prices; (iv) whether 'friendshoring' to allies (Japan, Korea, Netherlands) is a sufficient substitute. • Judgement: concentration is a material macro risk that justifies some but not complete reshoring, given the significant cost of full self-sufficiency.	(10)

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Question Number	Discuss the extent to which the US CHIPS Act and the EU Chips Act can be justified on grounds of economic efficiency, with reference to Extracts A and B.	Mark
	<i>Indicative content</i>	
1(c)	Knowledge/understanding 2, Application 2, Analysis 4, Evaluation 4 • Knowledge: productive, allocative, dynamic efficiency; positive production externalities (resilience, knowledge spillovers); infant-industry argument; public-good aspects of national defence; government failure — rent capture, subsidy race, distortion of comparative advantage. • Extract B — US CHIPS Act \$52bn + \$24bn tax credits triggering \$450bn+ private commitments; EU Chips Act €43bn; China Big Fund III \$47.5bn. Figure 2 — announced capex: US \$223bn, China \$142bn, Taiwan \$148bn. • Efficiency arguments FOR: (i) positive production externality — resilience and national security are public goods and under-supplied by the market; (ii) knowledge spillovers — semiconductor R&D; has large horizontal spillovers; (iii) coordination failures — infrastructure, talent pipelines, specialised suppliers require coordinated investment; (iv) infant-industry case for revived US capacity; (v) other countries subsidise — level playing field argument. • Efficiency arguments AGAINST: (i) departure from comparative advantage — TSMC Arizona fab is ~30% more expensive per wafer than Taiwan; (ii) subsidy race — the four major jurisdictions together subsidise \$160bn+; (iii) rent capture — public funds flow to profitable firms' shareholders; (iv) crowding out of private investment; (v) opportunity cost of public funds (healthcare, education, climate). • Evaluation: depends on (i) size of resilience externality — plausibly very large given the \$1tr global market; (ii) subsidy design (conditionality, clawbacks, export-control compliance); (iii) international coordination to avoid pure subsidy race; (iv) counterfactual — without the Act, US leading-edge share would likely have fallen to near zero. • Judgement: on narrow productive-efficiency grounds the Act is distortive, but given the scale of the positive externality and other jurisdictions' behaviour, it is plausibly welfare-improving — conditional on design.	(12)

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Question Number	Using Extracts B, C and D and your own knowledge, evaluate whether the ongoing 'technology decoupling' between the United States and China is likely to accelerate or hinder global economic growth over the next decade.	Mark
	<i>Indicative content</i>	
1(d)	Knowledge/understanding 4, Application 4, Analysis 6, Evaluation 6 • Knowledge: comparative advantage and gains from trade; scale economies and learning; specialisation vs duplication; R&D; spillovers; investment and capital deepening; global TFP growth; geopolitical risk premium; fiscal crowding-out. • Application: Extract B — successive export controls (2022, 2023, 2024); Extract C — Taiwan concentration and cyclicalities (DRAM prices –60%); Extract D — Nvidia revenue \$27bn → \$125bn+; 'compliant' variants A800/H20 subsequently restricted. Figure 1 — China's share of production rose from 0% (1990) to 19% (2024). Figure 2 — China announced \$142bn capex vs US \$223bn. Figure 3 — Nvidia R&D; \$5.3bn (FY22) → \$12bn (FY25). • Macro diagram: AD/AS showing higher investment shifting SRAS/LRAS right in advanced economies vs a negative supply-side shock from fragmentation; or PPF showing loss of world output from duplication. • Growth-accelerating channels: (i) massive investment stimulus — trillions in capital formation globally; multiplier and accelerator effects; (ii) competing R&D; effort plausibly accelerates innovation (space-race logic); (iii) reshoring creates middle-income advanced-economy manufacturing jobs; (iv) reduces catastrophic tail risk of supply-chain disruption — lower risk premia on investment; (v) government-led investment can correct market under-provision of resilience. • Growth-hindering channels: (i) loss of efficiency from abandoning comparative advantage; IMF estimates fragmentation cost at 0.2–7% of global GDP; (ii) reduced diffusion of technology to China slows productivity in the world's second-largest economy; (iii) escalating export-control uncertainty deters private investment; (iv) compliant-variant cycle (A800 → H20 → further limits) imposes redesign costs; (v) fragmented R&D; — duplication rather than complementary specialisation; (vi) commodity/supply-chain cost-push inflation; (vii) crowding out of other public investment. • Synoptic micro + macro: micro — some firms strengthened (TSMC, Samsung, Nvidia in Western markets; SMIC and Huawei in China), others weakened (Nvidia in China, ASML's China revenue). Macro — net effect depends on whether additional investment (demand-side) outweighs efficiency loss (supply-side); medium-run investment boom dominates, long-run efficiency loss accumulates unless R&D; effort genuinely accelerates innovation. • Evaluation: depends on (i) completeness of decoupling — partial decoupling at the frontier with continued cooperation on mature nodes is more benign than full bifurcation; (ii) response of non-aligned economies (India, Vietnam, Mexico); (iii) AI demand (Extract D) — may drive productivity gains offsetting fragmentation losses; (iv) crisis risk — a Taiwan contingency dominates all other factors. • A nuanced judgement — e.g. 'growth-positive in short run via investment but growth-negative in long run unless offset by AI productivity; balance depends on policy design and non-aligned-economy behaviour' — required for Level 4.	(20)

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Level 3	10–15	Accurate knowledge and understanding, supported by use of relevant evidence to support the argument, developed chains of reasoning, showing understanding of connections between causes and consequences. Arguments are well developed and competing arguments are present although this may lack balance. A conclusion may be attempted but may not show awareness of the significance of competing arguments.
Level 4	16–20	Accurate knowledge and understanding, supported throughout by use of relevant evidence which is well chosen and fully integrated to support the argument, well developed and logical, coherent chains of reasoning, showing full understanding of the question. Arguments are fully developed and evaluated. A full awareness of the validity and significance of competing arguments, leading to nuanced and balanced comparisons, judgements or conclusions.

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Section B — Question 2

Case-study context: generative AI — foundation models, Nvidia, labour market and regulation.

Question Number	Using Figure 3 (Extract D) and Figure 4, calculate the percentage-point change in Nvidia's gross margin between FY23 and FY24, and Nvidia's estimated share of 2024 AI chip sales. Discuss what these data suggest about Nvidia's degree of market power.	Mark
	<i>Indicative content</i>	
2(a)	Knowledge/understanding 2, Application 2, Analysis 2, Evaluation 2 - Gross margin change: $72.7 - 56.9 = +15.8$ percentage points. Accept also the relative change $(72.7/56.9 - 1) \approx +27.8\%$ if clearly explained. - Nvidia's share of 2024 AI chip sales: approximate annualised Nvidia AI chip revenue $\approx \\$100\text{bn}$ (midpoint of FY24 $\\$60.9\text{bn}$ and FY25 est. $\\$125\text{bn}$). Total market $\\$145\text{bn}$ (Figure 4). Share $\approx 100/145 \approx 69\%$. Accept a range 60–85% with working. - Market-power indicators: gross margin rose from 56.9% to 72.7% — an extraordinary expansion consistent with pricing power ($P \gg MC$). Combined with $\sim 69\%$ market share, this is strong evidence of substantial market power. - Demand is highly inelastic — no close substitutes at the frontier, CUDA software lock-in raises switching costs. Supply is constrained (TSMC advanced nodes). - Operating profit rose from $\\$4.2\text{bn}$ (FY23) to $\\$32\text{bn}$ (FY24); market cap from $\\$370\text{bn}$ to $\\$1.22\text{tr}$ — financial market expects rent capture to continue. - Evaluation: durability depends on (i) DeepSeek (Extract H) showed lower-cost inference is possible; (ii) custom silicon from hyperscalers (Google TPUs, Amazon Trainium) competes on inference; (iii) AMD MI300 and Intel Gaudi compete directly; (iv) CUDA lock-in may erode as open alternatives mature; (v) antitrust attention. - Judgement: Nvidia exhibits substantial market power in the short run; long-run sustainability much less certain.	(8)

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Question Number	With reference to Extract F, assess the likely macroeconomic consequences of the rise in AI-related data-centre electricity demand for the economies that host major data centres.	Mark
	<i>Indicative content</i>	
2(b)	Knowledge/understanding 2, Application 2, Analysis 3, Evaluation 3 • Knowledge: aggregate demand (investment, government spending), aggregate supply shifts, capacity utilisation, inflation, crowding out, regional multiplier, externalities (carbon), public goods (grid infrastructure). • Extract F: global data-centre electricity could double by 2026; hyperscalers have signed 30 GW of low-carbon contracts, including nuclear restart (Three Mile Island). Figure 4 — hyperscaler capex \$128bn → \$235bn over three years. • Positive macro effects: (i) investment boom — \$235bn hyperscaler capex directly raises AD; (ii) regional multiplier — DC construction employs thousands locally; (iii) anchor demand for clean power reduces financing costs for decarbonisation; (iv) productivity spillovers from cloud computing across the wider economy. • Negative macro effects: (i) pressure on electricity prices — doubling of DC demand can crowd out other users (Ireland has imposed a moratorium on new DCs in Dublin); (ii) cost-push inflationary pressure via higher business electricity prices; (iii) local externalities — water use, grid strain; (iv) concentration in a few regions amplifies local pressures; (v) if supply expansion relies on new nuclear (5–10+ yr lead time), short-run demand must be met by gas, undermining climate targets. • Evaluation: depends on (i) pace of supply expansion vs demand; (ii) location — economies with abundant renewable potential benefit more than constrained grids; (iii) productivity payoff — if Goldman's 1.5pp TFP gain materialises, the cost is trivially small relative to the benefit; (iv) regulatory response (siting policies, capacity payments). • Judgement: economies with surplus clean power and investment-friendly regulation are likely net beneficiaries; those with constrained grids face macro-inflationary pressure in the short run.	(10)

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Question Number	With reference to Extract G, discuss the likely impact of widespread adoption of generative AI on the labour market and income distribution in advanced economies.	Mark
	<i>Indicative content</i>	
2(c)	Knowledge/understanding 2, Application 2, Analysis 4, Evaluation 4 • Technology-driven labour effects: substitution (jobs replaced), complementarity (productivity enhanced), creation (new tasks). Skill-biased vs routine-biased technical change. Polarisation hypothesis. Factor shares (capital vs labour). • Extract G: Goldman 1.5pp annual productivity; MIT/Princeton task-level gains of 14–56%; IMF — 40% of global jobs exposed, 60% in advanced economies, higher exposure for women and higher earners. • Labour-market effects: (i) substitution in cognitive routine tasks (customer service, basic coding, legal review, copywriting); (ii) complementarity — AI-augmented workers become more productive (a programmer with Copilot produces more per hour); (iii) creation of new roles (prompt engineering, AI safety, data curation, AI-ethics); (iv) frictional/structural unemployment during transition; (v) value of generic cognitive skill falls; value of directing AI rises. • Income-distribution effects: (i) unlike previous waves, generative AI exposes higher-earning white-collar work — top-decile labour share may fall, but capital share rises — net effect on inequality ambiguous; (ii) gender dimension — IMF notes female-dominated roles have higher exposure; (iii) between-country inequality — advanced economies more exposed; (iv) factor shares — returns to AI capital (Nvidia gross margin 72%+) may shift factor income shares. • Evaluation: depends on (i) pace of capability growth — fast diffusion amplifies displacement; (ii) complementarity vs substitution balance; (iii) policy response — reskilling investment, social safety net, capital taxation; (iv) competition in AI markets — more competitive supply passes more surplus to labour/consumers; (v) new-job creation rate — historically technology creates more jobs than it destroys, but transitions are painful. • Judgement: the short-run vs long-run asymmetry, distributional complexity, and policy-dependence are critical; a nuanced answer is rewarded at Level 4.	(12)

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Question Number	Evaluate the appropriate role for governments in regulating the market for frontier AI foundation models, using evidence from Extracts E to H. Your answer should consider both microeconomic efficiency and wider macroeconomic consequences.	Mark
	<i>Indicative content</i>	

2(d)	Knowledge/understanding 4, Application 4, Analysis 6, Evaluation 6 • Knowledge: market failure — externalities (mis/disinformation, bias, safety), public goods (AI safety research), concentration/monopoly, asymmetric information. Government failure — regulatory capture, compliance cost, innovation chill, extraterritorial spillovers. Dynamic vs static efficiency; precautionary principle. • Application: Extract E — winner-take-all stack with hyperscaler–AI-firm partnerships; Extract F — training costs (\$100m → potentially \$10bn), energy demand; Extract G — EU AI Act with 7% turnover fines, US executive orders; Extract H — Microsoft–OpenAI \$13bn, Amazon–Google–Anthropic up to \$8bn; DeepSeek open-weight release (\$600bn market-cap wipe); antitrust investigations. • Micro case FOR regulation: (i) externalities — misinformation, discriminatory outputs, cyber-offensive use, concentration in critical infrastructure; (ii) asymmetric information — users cannot evaluate model safety, so disclosure rules help; (iii) concentration — hyperscaler partnerships may foreclose competing developers from cloud distribution; (iv) public-goods case for government support of AI safety research; (v) consumer protection against misleading AI content. • Micro case AGAINST: (i) regulatory cost disadvantages smaller developers (Mistral, open-source) relative to hyperscaler-backed incumbents — entrenching concentration; (ii) prescriptive regulation stifles dynamic efficiency; (iii) hard to specify regulation for rapidly evolving technology; (iv) DeepSeek episode shows frontier capability may emerge at far lower cost than assumed — reason to hesitate before regulating on winner-take-all assumptions; (v) regulatory capture — major labs may shape rules to raise rivals' costs. • Macro case FOR: (i) mitigates labour-market disruption via impact assessments / phased deployment; (ii) redistributive measures (data-worker rights, training levies) moderate inequality growth; (iii) supports financial stability by constraining AI use in high-risk applications; (iv) reduces geopolitical instability (bioweapon assistance, disinformation). • Macro case AGAINST: (i) slower adoption reduces productivity dividend — if Goldman's 1.5pp TFP estimate is correct, a one-year delay costs trillions of dollars of global output; (ii) regulatory fragmentation — EU AI Act, US executive orders, China registration — shrinks global addressable market; (iii) innovation flight to lighter-touch jurisdictions; (iv) hyperscaler capex (\$235bn in 2024) could be deterred if regulatory uncertainty raises required returns. • Synoptic integration: micro and macro considerations cut in different directions for different instruments. Antitrust (e.g. blocking Microsoft's equity link to OpenAI) addresses micro concentration, limited macro cost. Safety/disclosure obligations (EU AI Act) address micro externalities with macro productivity cost only if poorly calibrated. Ex-ante licensing of frontier training imposes large macro costs; ex-post liability imposes lower macro cost but may under-deter. • Evaluation: depends on (i) empirical magnitude of AI harms — uncertain, arguing for an adaptive framework; (ii) international coordination vs fragmentation; (iii) specific instrument — competition enforcement vs safety disclosure vs licensing; (iv) speed of adaptation; (v) allocation of cost (developers vs deployers); (vi) treatment of open-weight models. • A nuanced judgement — e.g. 'intervention appropriate for clearly-identified externalities (competition, safety disclosure, high-risk uses) but should avoid prescriptive ex-ante licensing, preserving the dynamic efficiency on which the macroeconomic productivity dividend depends' — required for Level 4.	(20)
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